Sanjay Kumar S S

Website: sanjaykumar.uk
LinkedIn: linkedin.com/in/sanjay-kumar-ss
Email : contact@sanjaykumar.uk
Mobile : +44 7344754782
Github: github.com/SS-Sanjay-Kumar

Education

- **University of Leeds** — Leeds, England
  Master of Science in Advance Computer Science (AI) — *Sept. 2026 – Sept. 2027 (expected)*

Experience

- **Indominus Labs** — Chennai, Tamil Nadu, India
  AI Engineer Intern — *Nov 2025 - May 2026 (7 mos)*
  - Engineered high-throughput backend microservices and async AI pipelines, maintaining sub -100ms API response latency across production endpoints.
  - Developed an AI-powered conversational chatbot leveraging Retrieval-Augmented Generation (RAG) and Text-to-SQL to process complex medical domain knowledge.
  - Architected end-to-end CI/CD pipelines and automated deployments to AWS EC2 instances, reducing manual deployment overhead.
  - Integrated React Quill rich-text components into internal admin workflows, reducing content entry time by 30% for internal users.
- **Cura(Auralis)** — Singapore
  Full Stack Developer Intern (Remote) — *May 2025 - Sept 2025 (7 mos)*
  - Designed and scaled core backend architecture for a fast-paced medical startup, directly supporting the platform's transition into a successful seed funding round.
  - Engineered an AI-powered medical chatbot using RAG and vector embeddings to search 10,000+ clinical documents in under 200ms.
  - Optimized relational database schemas and API query endpoints, slashing p99 latency by 45% under concurrent traffic loads
  - Translated complex clinical domain research into automated software workflows, ensuring high system uptime and data accuracy.

Projects

- **Vigilis** — Link
  An AI-driven, High-Throughput Log Analysis and Security Intelligence Engine.
  - Engineered a high-throughput Go and Redis log ingestion engine capable of processing 10,000+ logs/sec at sub-5ms P95 latency by prioritizing buffered in-memory channels over disk-backed persistence.
  - Integrated an unsupervised PyTorch deep autoencoder that evaluates raw log sequence embeddings across 11.1M+ HDFS records, flagging security anomalies based purely on reconstruction loss (MSE) thresholds.
  - Architected a low-latency Server-Sent Events (SSE) streaming endpoint to broadcast neural network detection alerts in real time, choosing unidirectional HTTP streaming over WebSockets to minimize connection state overhead.
  - Optimized PostgreSQL dataset sampling across 11.1M+ rows using primary key indexed sequence offsets instead of ORDER BY random(), slashing query execution latency from 1.2s to under 5ms.
- **Mentis** — Link
  A real-time collaborative coding interview platform with video calls, live code editor and chat.
  - Architected a real-time collaborative code editor and multi-party video calling system using WebRTC and Stream SDKs to achieve sub-50ms sync latency.
  - Engineered low-latency WebSockets and MongoDB change streams to power instantaneous chat and live code execution state synchronization across active rooms.
  - Implemented secure JWT authentication and route guard middleware via Clerk, enabling isolated candidate session management with zero auth overhead.

Technical Skills

Languages: Go, Python, JavaScript, SQL
Backend & AI: FastAPI, Node.js, Express, PyTorch, TensorFlow, LangChain, RAG, REST APIs, SSE, WebSockets
Databases & Caching: PostgreSQL, MongoDB, Redis, Distributed Caching, Schema Optimization
DevOps & Infrastructure: Docker, AWS (EC2), GitHub Actions CI/CD, Prometheus, Linux, Git